

Dhruv Rastogi

AI / Backend Engineer — Applied ML · RAG · Distributed Systems

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SUMMARY

1.5+ years of hands-on production software engineering across three internships, building across the AI application stack from backend services in Java and Spring Boot to applied-ML products. Built AI-Court, a deployed legal-outcome ML system (scikit-learn classifier, 91.8% accuracy on 10,838 cases, with semantic precedent retrieval and data-drift monitoring) and DhrLang, a statically-typed language with a full compiler, JVM + EVM backends, and a formal verifier (1,400+ tests in CI). Currently interning at MAQ Software through 31 October 2026 on 500K+ records-a-day Azure data pipelines.

SKILLS

AI / ML: Applied machine learning (scikit-learn — TF-IDF, ensemble classifiers, SMOTE), Retrieval-Augmented Generation (RAG) & semantic search (Sentence-Transformers), MLOps (Prometheus metrics, data-drift monitoring, model governance), model evaluation & explainability, prompt design

Languages: Python, Java (primary), JavaScript / TypeScript, SQL, Bash

Backend: Spring Boot, REST APIs, microservices, authentication, role-based access control (RBAC), async / queue-based processing

Cloud & Data: Microsoft Azure (Functions, Fabric, Databricks, DevOps), Docker, Kubernetes, Terraform, MySQL, PostgreSQL, MongoDB, Redis, ETL pipelines

PROJECTS

AI-Court — ML legal-outcome platform

github.com/dhruv-15-03/AI-CourtRoom

A Python/Flask ML service that predicts Indian legal-case outcomes: a TF-IDF + boosted Random Forest classifier (91.8% test accuracy, 0.83 macro-F1 across 10,838 cases) with semantic precedent retrieval (Sentence-Transformers), confidence-based abstention, explainable top-feature output, and an MLOps spine (Prometheus metrics, data-drift monitoring, model governance). Java + React app shell; deployed on Render and Vercel.

DhrLang — statically-typed JVM + EVM language (v4.0.0)

github.com/dhruv-15-03/DhrLang

Compiler written end to end: lexer, parser, type checker, IR lowering, and three execution backends (AST, IR VM, JVM bytecode) plus an EVM/Solidity target and a from-scratch formal verifier (Fourier–Motzkin static analysis with SARIF Code Scanning output). Ships an LSP server and a published VS Code extension. 14 releases; 1,400+ tests.

boot-usage — Spring Boot Actuator starter (Apache-2.0)

github.com/dhruv-15-03/boot-usage

Drop-in Spring Boot starter exposing Actuator endpoints for runtime dependency analysis. Published as a reusable Java package.

EXPERIENCE

Associate Software Engineer Intern · MAQ Software

Nov 2025 – Present · ends 31 Oct 2026 · India

Java · Python · Azure Fabric · Databricks · Azure Functions · Terraform · Azure DevOps

- Build and run ETL pipelines on Azure Fabric and Databricks that process 500,000+ records a day at 99.5% accuracy, cutting manual data prep by about 60%.
- Own CI/CD for 10+ Azure Function Apps using Azure DevOps and Terraform, enabling zero-downtime releases and roughly 40% faster deployments.
- Improved alerting, monitoring, and reliability tuning for services running on a 99.9% uptime SLA.

Associate Software Engineer Intern · RecruitCRM

Jun 2025 – Nov 2025 · Remote

Java · Spring Boot · MySQL · REST APIs · Vue.js · RBAC

- Shipped 8+ REST APIs for CRM workflows, lead enrichment, and live data access, with 85% test coverage on the new code.
- Implemented authentication and role-based access control across the backend.
- Reduced API response times by 30% through query and execution-path optimization.

Software Engineer Intern · CEERAS

Jan 2025 – Jun 2025 · Remote

Java · Microservices · Docker · Kubernetes · Redis · GitHub Actions

- Built a distributed bidding system handling 10,000+ transactions a day at sub-second latency, backed by async processing and a Redis cache layer.
- Set up the Docker, Kubernetes, and GitHub Actions pipeline for build, test, and deployment.

EDUCATION

B.Tech, Computer Science & Engineering (AI)

June 2026

Meerut Institute of Engineering and Technology (MIET) / Dr. A.P.J. Abdul Kalam Technical University (AKTU), India · 8.6 / 10.

CERTIFICATIONS

Selected issuer-verified credentials — [SQL AI Developer Associate](#) · [Fabric Data Engineer Associate, DP-700](#) · [Azure Developer Associate, AZ-204](#) · [DevOps Engineer Expert, AZ-400](#)

Oracle Cloud Infrastructure 2025 — [Developer Professional](#) · [DevOps Professional](#) · [MongoDB Associate Developer](#) · [GitHub Copilot Certified](#)

ACHIEVEMENTS

LeetCode Knight (1,000+ problems solved) · Smart India Hackathon 2024 — Top 50 nationally · ISC Class XII — 96.75% (Rank 1 on campus)

Available in 2 weeks. Open to AI Engineer, Applied AI / LLM, and Backend roles — remote, hybrid, or relocation.